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ADDIS ABABA INSTITUTE OF TECHNOLOGY

SCHOOL OF INFORMATION TECHNOLOGY AND ENGINEERING

Department of [Name]

Title of the Thesis: Concise, Clear and Descriptive

A thesis submitted to the School of Information Technology and Engineering, Addis Ababa Institute of Technology, Addis Ababa University, in partial fulfillment of the requirements for the degree of Master of Science in [Program Name]

By

Full Name of the Student

Advisor: Dr. Full Name of the Advisor

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Addis Ababa, Ethiopia

November 2026

ADDIS ABABA UNIVERSITY

ADDIS ABABA INSTITUTE OF TECHNOLOGY

SCHOOL OF INFORMATION TECHNOLOGY AND ENGINEERING

Title of the Thesis: Concise, Clear and Descriptive

By

Full Name of the Student

Approval by Board of Examiners

This is to certify that the thesis prepared by Full Name of the Student, entitled "Title of the Thesis: Concise, Clear and Descriptive", and submitted in partial fulfillment of the requirements for the degree of Master of Science in [Program Name], complies with the regulations of the University and meets the accepted standards with respect to originality and quality.

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DECLARATION

I, the undersigned, declare that this thesis is my original work, that it has not been presented for a degree in this or any other university, and that all sources of material used for the thesis have been duly acknowledged.

	Name	**Signature**	**Date**
Student	Full Name of the Student	_____	_____

This thesis has been submitted for examination with my approval as university advisor.

	Name	**Signature**	**Date**
Advisor	Dr. Full Name of the Advisor	_____	_____
Co-advisor	Dr. Full Name of the Co-advisor	_____	_____

Place: Addis Ababa, Ethiopia
Date of submission: November 2026

To . . .

ABSTRACT

Guide: One page that tells the reader why the problem matters, what you did and how, what the main results were (with numbers), and what the work contributes. Write it last, and do not cite references in it.

Write the abstract here. Start with the problem and why it matters. Then describe the approach and datasets, give the main quantitative results, and end with the contribution of the work.

Keywords: keyword one, keyword two, keyword three, keyword four

ACKNOWLEDGMENT

Write your acknowledgment here.

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LIST OF ACRONYMS

AAiT	Addis Ababa Institute of Technology
AAU	Addis Ababa University
CNN	Convolutional Neural Network
ConvLSTM	Convolutional Long Short-Term Memory
GPU	Graphics Processing Unit
SiTE	School of Information Technology and Engineering

CHAPTER ONE

INTRODUCTION

1.1. Background

> **Guide:** Give the reader the facts they need to understand the topic and how it links to your research question. Start from the broad theme, then narrow down to the specific details that make your problem relevant. By the end, the need for the research should already be clear.

Write the background here. Cite sources as you go [1], and introduce acronyms with `\ac{...}`. For example, Convolutional Neural Network (CNN) is spelled out on first use and abbreviated afterwards: CNN.

1.2. Motivation

> **Guide:** Explain what motivated you to carry out this work.

Write the motivation here.

1.3. Statement of the Problem

> **Guide:** Define the problem in the literature, theory or practice that this thesis solves, clearly enough that the reader knows exactly what is being addressed. A good problem statement answers "Why does this research need to be conducted?"

Write the problem statement here.

1.4. Research Questions

> **Guide:** List the questions the thesis answers. Chapter 6 should answer each one explicitly.

This study answers the following research questions:

RQ1. First research question?

RQ2. Second research question?

RQ3. Third research question?

1.5. Objectives

Guide: State a general objective, and specific objectives that are simple, specific and follow from the problem statement.

1.5.1. General Objective

The general objective of this study is to . . .

1.5.2. Specific Objectives

To achieve the general objective, the study will:

- review . . . ;
- prepare . . . ;
- design and implement . . . ;
- evaluate . . . .

1.6. Research Methodology

Guide: A short overview of how the objectives were achieved: literature review, data, tools and evaluation. Keep it brief and point to Chapter 4 for the details.

1.6.1. Literature Review

Describe how the literature was reviewed.

1.6.2. Data Collection

Describe the dataset(s) used.

1.6.3. Tools and Techniques

List the main tools, frameworks and hardware.

1.6.4. Evaluation

Describe how the results were evaluated.

1.7. Scope and Limitations

Guide: State what the study covers, and what it does not cover and why.

Scope. Describe what the study covers.

Limitations. Describe what the study does not cover.

1.8. Contributions

Guide: List the concrete contributions of the thesis (new method, dataset, results, analysis) and how they extend existing knowledge.

The main contributions of this thesis are:

- . . . ;
- . . . .

1.9. Organization of the Thesis

The rest of this thesis is organized as follows. Chapter 2 reviews the background concepts and techniques. Chapter 3 discusses related work and identifies the research gap. Chapter 4 presents the proposed approach. Chapter 5 describes the experiments and discusses the results. Chapter 6 concludes the thesis and outlines future work.

CHAPTER TWO

LITERATURE REVIEW

Guide: Cover the concepts, theories and techniques the reader needs to understand the rest of the thesis. Organise the chapter by topic, cite the original sources, and keep it focused on what your work builds on.

Give a short introduction to the chapter here.

2.1. Overview of [Topic]

Write about the topic here. A book is cited like this **goodfellow2016deep**.

2.2. [Core Concept or Technique]

2.2.1. [Sub-topic]

Write about the sub-topic here.

2.2.2. [Sub-topic]

Write about the sub-topic here.

2.3. Datasets

Describe the publicly available datasets in the area.

2.4. Evaluation Metrics

Describe the metrics commonly used in the area [2].

2.5. Summary

Summarise the chapter in a short paragraph.

CHAPTER THREE

RELATED WORK

Guide: Discuss the studies closest to your work, grouped by approach. For each group, say what was done, how well it worked and what is still missing. End by stating the gap that this thesis fills, and show how your work is similar to, yet different from, previous work.

Give a short introduction to the chapter here.

3.1. [First Group of Approaches]

Discuss the first group of related works here [3].

3.2. [Second Group of Approaches]

Discuss the second group of related works here [4].

3.3. Summary and Research Gap

Every table must be referred to in the text. Table 3.1 summarises the related works discussed in this chapter.

Table 3.1: Summary of related works

Work	Approach	Dataset / Evaluation	Limitation / Gap
[1]	...	...	...
[3]	...	...	...
[4]	...	...	...

State the research gap that this thesis addresses.

CHAPTER FOUR

METHODOLOGY

> **Guide:** Describe what you built or did, in enough detail that someone else could reproduce it. Justify each design choice and relate it back to the research questions.

This chapter describes how the research questions of Section 1.4 were investigated. It first sets out the research design, which fixes every decision rule before the experiment that it governs (Section 4.1), and formalises the prediction task (Section 4.2). It then describes the eye-tracking data and their preparation (Section 4.3), the proposed model, SURGE-D, as it is implemented (Section 4.4), the components that were implemented and tested but not retained (Section 4.5), the training and inference procedures (Sections 4.6 and 4.7), and the evaluation metrics and statistical analysis (Section 4.8). The chapter closes with the ethical considerations of the work (Section 4.9). Every equation in this chapter corresponds to code in the accompanying repository; where the implementation departs from the original research proposal, the departure is stated.

4.1. Research Design

The study follows a design-science approach in which an artefact, a video saliency model, is built and then evaluated through controlled experiments. Two features of the design deserve emphasis, because they determine how the results in Chapter 5 may be read.

Pre-registered decision rules. The original proposal contained several components whose value was uncertain: the choice of video encoder, a "surprise" signal derived from the encoder's predictor, parameter-efficient adaptation of the encoder, static-image co-training, and others. Rather than include all of them and report the final configuration, each component was treated as a hypothesis with an explicit test. Before each experiment was run, its decision rule (the quantity to be measured, the data on which it is measured, and the threshold that decides the outcome) was written into a dated research log, and the rule was applied unchanged afterwards. Table 4.1 lists these rules. This procedure guards against two familiar sources of optimism in deep-learning research: choosing among variants after seeing test results, and adjusting thresholds after seeing which side of them a result falls on.

Separation of selection and testing. All model selection (the best training epoch, the output blur, every gate in Table 4.1) was performed on internal holdout sets carved out of the training data. Each test set was scored once per reported model, after all of that model's settings had

been fixed. For DHF1K, whose official test annotations are withheld by the dataset authors, the 100-video validation set serves as the test set and was never used for selection.

Table 4.1: Decision rules, each written before the experiment that it governs. Holdout CC is measured on the internal 60-video DHF1K holdout unless stated otherwise; outcomes are reported in Chapter 5

Decision	**Rule**
Evaluator validity	Re-scoring the same maps with an independent evaluator must agree within 0.002 (CC, SIM) and 0.01 (NSS, KLD)
Backbone (gate G1)	Highest holdout CC wins; the proposal's choice, V-JEPA 2.1, is kept if within 0.005 CC of the winner
Surprise (gate G2)	Enabled only if a surprise map beats both token atypicality and frame difference on NSS and has positive information gain over the centre prior
LoRA adaptation	Adopted if the three-seed mean holdout gain is at least +0.003 CC and LoRA wins in at least two of three seeds
Single-seed ablations	A difference is reported only if $\|\Delta\text{CC}\| \geq 0.003$ on the holdout
Claim over a baseline	The 95 % bootstrap interval of ΔCC over test videos must lie above zero
ViT-g backbone	Adopted if its holdout CC is at least 0.5833 (the ViT-L three-seed mean plus 0.005)
Training on 600 videos	Replaces the 540-video models if the ensemble's ΔCC interval is above zero and it is worse on at most two of six metrics
32-frame clips	Pursued if the holdout CC is at least 0.5904 (the 16-frame three-seed mean plus 0.005)
Audio fusion	"Helps" on a dataset only if the fold-mean ΔCC is at least +0.010 and matched audio beats rotated audio by at least 0.005
Published results	Comparisons with numbers from other evaluation code treat margins below 0.01 CC as ties

Unit of analysis. Saliency metrics are computed per frame and averaged within each video; the video is the unit of analysis, and uncertainty is estimated by resampling videos (Section 4.8.2). Ablations and gates with a single training seed are interpreted against the seed-to-seed spread observed for the main model, which motivates the 0.003 and 0.005 margins in Table 4.1.

4.2. Problem Formulation

Let a video be a sequence of frames $I_1, \ldots, I_N$ with $I_n \in \mathbb{R}^{H \times W \times 3}$. For each frame, an eye-tracking dataset provides a binary fixation map $F_n \in \{0, 1\}^{H \times W}$, marking the pixels fixated by at least one observer, and a continuous fixation density Y_n, obtained by blurring the fixations with a Gaussian kernel. Video saliency prediction is the task of estimating, for every frame, a probability distribution $\hat{S}_n$ over pixel locations,

$$\hat{S}_n(p) \geq 0, \qquad \sum_p \hat{S}_n(p) = 1, \tag{4.1}$$

that approximates where observers look in frame n. Treating saliency as a distribution, rather than as an unnormalised score map, is the probabilistic view advocated by Kümmerer et al. [5], [6]; it makes the training objective and the distribution-based metrics well defined.

The model sees a temporal window rather than a single frame. A clip of $T = 16$ frames is sampled with a stride of $s = 2$, so that the sampled indices are $n_k = n_0 + ks$ for $k = 0, \ldots, T-1$; at the 30 frames-per-second rate of DHF1K the clip spans 31 original frames, or about one second. Indices that fall outside the video are clamped to its first or last frame. The model predicts maps for all T sampled frames but only the central $T/2$ positions,

$$O = \{T/4, \ldots, 3T/4 - 1\} = \{4, \ldots, 11\}, \tag{4.2}$$

are used for training and inference. Each prediction therefore has at least eight original frames (about 0.27 s) of context on either side. The formulation is thus *non-causal*: a frame's map depends on frames that follow it. This suits offline applications such as video compression or summarisation but not streaming use, and it differs from methods that predict the last frame of a clip from past frames only, such as SalFoM [7]. The difference is returned to when results are compared in Chapter 5.

4.3. Datasets

4.3.1. Data Collection

No new eye-tracking data were collected. The study uses ten public datasets, summarised in Table 4.2. All were read from their original distributions; the splits and holdout lists used here are released with the code.

DHF1K [3], [8] is the primary benchmark. Its 600 training videos were split once, with a fixed random seed, into 540 training videos and a 60-video holdout used for all model selection. The official validation set (videos 601–700, 60,273 frames) is the test set of this thesis. Predictions for the 300 official test videos were generated with the final model and packaged in the format required by the dataset authors; their scores were not available when this thesis was written and are not reported.

UCF Sports [9] and Hollywood-2 [11] are action-recognition datasets for which Mathe and Sminchisescu [10] recorded the gaze of observers performing action recognition. For UCF Sports the standard 103/47 split was used, with ten training videos held out. The Mathe and Sminchisescu ground truth for Hollywood-2 could not be obtained during the project; instead, the independently recorded and openly licensed gaze data of Vig, Dorr and Cox [12] were used[1]. These data come from fewer observers and cover whole clips rather than the shot-segmented clips used in the saliency literature, so results on them are reported under the

[1] Figshare item 93951, released under the CC BY licence; every file was checked against its published checksum.

Table 4.2: Eye-tracking datasets used in this thesis. "Holdout" is the internal selection set drawn from the training part; the test part is scored once per model

Dataset	Content and viewing condition	Split used (train / holdout / test)
DHF1K [3], [8]	1,000 YouTube videos, 640 × 360 at 30 fps, free viewing, 17 observers; official split 600 / 100 / 300 (test labels withheld)	540 / 60 / 100 (official validation set as test)
UCF Sports [9], [10]	150 sports videos; gaze recorded during action recognition; standard split 103 / 47	93 / 10 / 47
Hollywood-2 [11] with the gaze of Vig et al. [12]	1,707 movie clips (823 / 884); four observers who watched every clip plus a pooled fifth observer, recorded during action recognition	741 / 82 / 884
DIEM [13]	84 videos of many genres; audio present	one standard split, 64 training and 17 test videos
Coutrot1 [14], [15]	60 clips in four visual categories	three folds each, every video tested once [16]; holdout = 10 % of each training fold
Coutrot2 [14], [15]	15 clips of four people in a meeting	
ETMD [17], [18]	12 clips from six Hollywood movies	
AVAD [19]	45 short audio-visual clips	
SumMe [18], [20]	25 user-made videos	
SALICON [21]	10,000 static training images	co-training ablation only

separate name "Hollywood-2 (Vig gaze)" and are not compared with published Hollywood-2 numbers.

The six audio-visual datasets (DIEM [13], Coutrot1 and Coutrot2 [14], [15], ETMD [17], AVAD [19] and SumMe [20]) were used in the preprocessed form and with the fold lists released by Tsiami et al. [16], whose protocol has become the standard for these data: DIEM has one fixed split, and each of the other five datasets has three folds in which every video appears in exactly one test set. The gaze data for ETMD and SumMe were recorded by Tsiami et al. [18]. SURGE-D is a visual model; the soundtracks were used only in the audio experiment of Section 4.5.3. SALICON [21], a dataset of static images with mouse-tracking saliency, was used only in the static co-training ablation.

4.3.2. Preprocessing

Frames. Frames were read as distributed (DHF1K, UCF Sports and the STAViS releases) or decoded once with OpenCV (Hollywood-2, 487,556 frames). For the model, every frame is resized to 256 × 256 pixels with area interpolation, which changes the aspect ratio of 16:9 video, and normalised with the ImageNet channel means and standard deviations [22]. Squashing was chosen over cropping because cropping removes image borders and alters the centre-bias statistics of the fixation maps; its effect on accuracy was not ablated.

Training targets. For each target frame, the ground-truth density is resized with area interpolation to the 64×64 output grid and normalised to sum to one, and the fixated pixels of the binary fixation map are counted on the same grid. The resulting pair (Y_t, F_t) is used by the loss (Section 4.6). Evaluation, in contrast, always takes place at the native resolution of the ground truth (Section 4.8).

Frames without fixations. In the STAViS releases and in the Hollywood-2 (Vig gaze) maps, some frames have no recorded fixation, for example during blinks. These frames are excluded from scoring. DHF1K and UCF Sports contain no such frames.

Hollywood-2 (Vig gaze) maps. The Vig gaze recordings provide raw 1000 Hz samples with confidence values and saccade annotations. A frame f of a clip with frame rate r was taken to cover the interval $[(f-1)/r,\ f/r)$. For each observer and frame, the samples with confidence 1.0 that do not fall inside an annotated saccade were collected. If any remained, the observer contributed one fixation point, the median of those samples, rounded and clipped to the frame. The binary fixation map thus has at most five points per frame. The density map is this binary map blurred with a Gaussian of $\sigma = 20$ pixels, following the DHF1K recipe [3]. Of the 487,556 frames, 99.8 % received at least one point, with 4.53 points per frame on average. For model selection, 82 of the 823 training clips were held out with a fixed seed.

Audio features. For the audio experiment only, each soundtrack of the STAViS release was mixed to mono and resampled to 16 kHz, and the frozen Audio Spectrogram Transformer [23] fine-tuned on AudioSet [24] was applied to consecutive 10.24 s chunks[2]. Averaging the patch tokens over the twelve frequency rows of the patch grid gives one 768-dimensional vector per time column, that is, roughly one every 0.1 s. The feature of video frame k (centred at time $(k-\frac{1}{2})/r$) is the mean of the columns whose centres lie within ±0.5 s of that time. Videos without a soundtrack receive zero features.

4.4. Proposed Architecture

SURGE-D consists of a frozen self-supervised video encoder and a light trainable decoder (Figure 4.1). The decoder reads four intermediate layers of the encoder, rebuilds a multi-scale feature pyramid, decodes it top-down with factorised spatio-temporal convolutions, and adds a learned centre prior before normalising each output frame into a probability distribution. Only the decoder and the prior are trained: 6.99 million parameters with the ViT-L encoder and 7.39 million with ViT-g.

[2]Checkpoint `MIT/ast-finetuned-audioset-10-10-0.4593`, BSD-3-Clause licence.

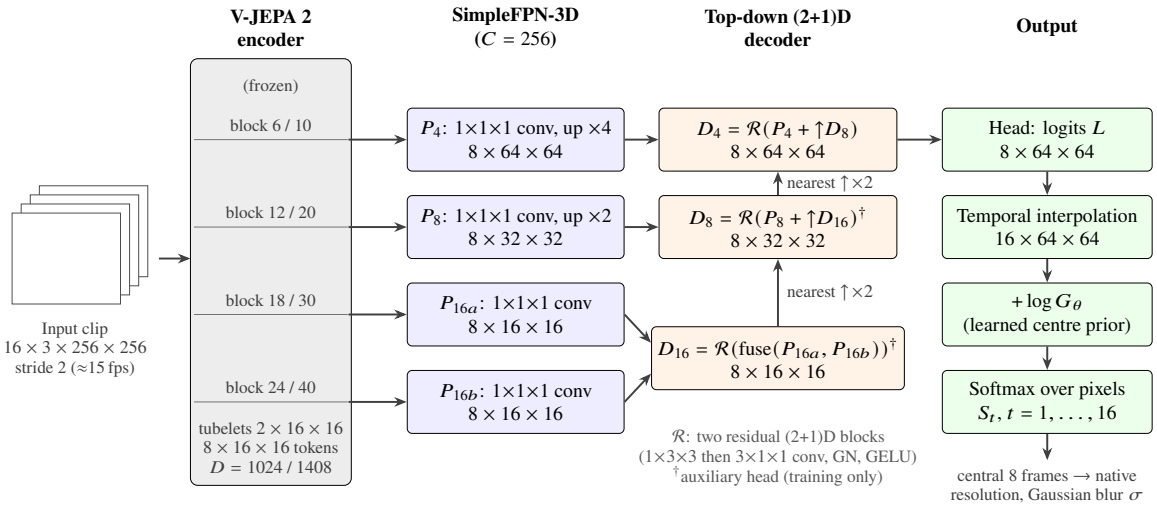


Figure 4.1: Architecture of SURGE-D. A clip of 16 frames passes through the frozen V-JEPA 2 encoder; four intermediate layers (blocks 6/12/18/24 of ViT-L or 10/20/30/40 of ViT-g) feed a three-dimensional simple feature pyramid, which a top-down (2+1)D decoder turns into logits at 64 × 64 resolution for eight tubelet steps. After temporal interpolation to 16 frames and the addition of a learned centre prior, a softmax over pixels yields one saliency distribution per frame; the central eight frames are kept. Tensor sizes are given as time × height × width

4.4.1. Frozen Video Encoder

The encoder is a Vision Transformer [25] trained with the video joint-embedding predictive architecture (V-JEPA) objective of Bardes et al. [26]. In this objective, which extends the image model I-JEPA [27] to video, parts of a clip are masked and a predictor network is trained to predict the *representations* of the masked regions, produced by an exponential-moving-average target encoder, from the representations of the visible ones. Because the loss is computed in representation space rather than pixel space, the encoder is not required to model unpredictable pixel detail, which Bardes et al. argue leads to features that transfer well with a frozen backbone [26].

The checkpoints used are the publicly released V-JEPA 2 models[3], which follow this objective: as Mur-Labadia et al. describe it, an L_1 loss between the predictor's outputs and the target encoder's representations is applied to the masked tokens only [29]. The same study reports that this choice leaves the visible tokens without a direct training signal and that frozen V-JEPA 2 features consequently carry little local structure, reaching only 22.2 mIoU in linear-probe semantic segmentation on ADE20K. Whether such features suit a dense task such as saliency prediction was therefore an open question, which gate G1 addresses (Section 5.3). Two sizes were used. ViT-L/16 has 24 transformer blocks of width $D = 1024$ and 303.9 million encoder parameters; ViT-g/16 has 40 blocks of width $D = 1408$ and 1,012.2

[3] `facebook/vjepa2-vitl-fpc64-256` and `facebook/vjepa2-vitg-fpc64-256`, released under the MIT licence through the Hugging Face hub and loaded with the Transformers library [28]. The technical report that introduced them had not appeared in a peer-reviewed venue when this thesis was written and is therefore not cited; their properties are taken from the checkpoints themselves and from the peer-reviewed follow-up study of Mur-Labadia et al. [29].

million parameters (both counts measured on the loaded checkpoints, excluding the predictor). Both embed the video in *tubelets* of $2 \times 16 \times 16$ pixels, so a $16 \times 256 \times 256$ clip becomes $8 \times 16 \times 16 = 2{,}048$ tokens.

Let $h^{(\ell)} \in \mathbb{R}^{2048 \times D}$ denote the output of block ℓ. Four blocks at equal fractions of the depth are tapped, $\ell \in \{6, 12, 18, 24\}$ for ViT-L and $\ell \in \{10, 20, 30, 40\}$ for ViT-g, and each tap is reshaped into a feature volume of size $D \times 8 \times 16 \times 16$. Intermediate layers are used because dense prediction benefits from features at several levels of abstraction [30], [31]. The encoder is frozen: its weights are not updated and no gradient is propagated into it. Freezing was the outcome of an experiment rather than an assumption; the alternative, low-rank adaptation of the encoder, is described in Section 4.5.2 and evaluated in Chapter 5.

4.4.2. Three-Dimensional Simple Feature Pyramid

A plain Vision Transformer produces features at a single resolution. Li et al. [32] showed that a feature pyramid for dense prediction can be built from such a backbone by simple up- and down-sampling of its features. SURGE-D uses a three-dimensional variant built from the four taps rather than from the last layer only. With $\phi(\cdot)$ denoting group normalisation [33] followed by a GELU non-linearity, W_ℓ a $1 \times 1 \times 1$ convolution from D to $C = 256$ channels, and $U(\cdot)$ a spatial $\times 2$ transposed convolution (kernel and stride $1 \times 2 \times 2$) followed by ϕ, the pyramid levels are

$$\begin{aligned} P_4 &= U\Big(U\Big(\phi(W_1 h^{(\ell_1)})\Big)\Big) \in \mathbb{R}^{C \times 8 \times 64 \times 64}, & P_8 &= U\Big(\phi(W_2 h^{(\ell_2)})\Big) \in \mathbb{R}^{C \times 8 \times 32 \times 32}, \\ P_{16a} &= \phi(W_3 h^{(\ell_3)}) \in \mathbb{R}^{C \times 8 \times 16 \times 16}, & P_{16b} &= \phi(W_4 h^{(\ell_4)}) \in \mathbb{R}^{C \times 8 \times 16 \times 16}. \end{aligned} \tag{4.3}$$

The shallowest tap feeds the finest level, as in standard feature pyramids, and the two deepest taps both remain at the token resolution of $1/16$. The temporal resolution of eight tubelet steps is preserved throughout.

4.4.3. Top-Down (2+1)D Decoder

The decoder refines the pyramid from coarse to fine. Its building block is a residual block that factorises a three-dimensional convolution into a spatial and a temporal part, in the manner of R(2+1)D [34],

$$\mathcal{B}(x) = \mathrm{GELU}\Big(x + \mathrm{GN}\big(\mathrm{Conv}_{3\times1\times1}\big(\mathrm{GELU}(\mathrm{GN}(\mathrm{Conv}_{1\times3\times3}(x)))\big)\big)\Big), \tag{4.4}$$

where the subscripts give the kernel size as time $\times$ height $\times$ width and GN is group normalisation. Let $\mathcal{R}$ denote two such blocks in sequence and $\uparrow$ nearest-neighbour spatial upsampling by a

factor of two. The decoder computes

$$\begin{aligned} D_{16} &= \mathcal{R}\big(\phi(\mathrm{Conv}_{1\times1\times1}[P_{16a}; P_{16b}])\big), \\ D_8 &= \mathcal{R}\big(P_8 + \uparrow D_{16}\big), \\ D_4 &= \mathcal{R}\big(P_4 + \uparrow D_8\big), \end{aligned} \tag{4.5}$$

where $[\cdot;\cdot]$ concatenates channels. A head with a $1 \times 3 \times 3$ convolution to 128 channels, a GELU and a $1 \times 1 \times 1$ convolution to one channel turns D_4 into logits $L \in \mathbb{R}^{8\times64\times64}$. During training, two auxiliary heads ($1 \times 1 \times 1$ convolutions) produce additional logits from D_{16} and D_8; they are upsampled to 64×64 and supervised with the same targets, a form of deep supervision [35]. The auxiliary heads are not used at inference.

4.4.4. Temporal Upsampling, Centre Prior and Output

The logits have one time step per tubelet. They are interpolated trilinearly to the $T = 16$ sampled frames, giving L_t for $t = 1, \ldots, T$.

Human fixations in videos are biased towards the centre of the screen, partly because of how scenes are framed and partly because of viewing strategy [36], [37]. SURGE-D models this bias with a learned, input-independent prior. With pixel positions $p \in [0, 1]^2$ in normalised coordinates, the prior is a mixture of $K = 8$ Gaussians with diagonal covariance,

$$\log G_\theta(p) = \alpha \left[\log \sum_{k=1}^{K} \pi_k \, \mathcal{N}\Big(p; \mu_k, \mathrm{diag}(\sigma_k^2)\Big) - \log \sum_{p'} \sum_{k=1}^{K} \pi_k \, \mathcal{N}\Big(p'; \mu_k, \mathrm{diag}(\sigma_k^2)\Big) \right], \tag{4.6}$$

where the mixture weights $\pi = \mathrm{softmax}(w)$, the means μ_k, the scales σ_k and the strength α are learned (41 parameters in total). The means are initialised on a grid in the central $[0.35, 0.65]^2$ region, the scales at 0.25 and the strength at one. The prediction for frame t is

$$S_t(p) = \frac{\exp\big(L_t(p) + \log G_\theta(p)\big)}{\sum_{p'} \exp\big(L_t(p') + \log G_\theta(p')\big)}, \tag{4.7}$$

a softmax over the 64×64 positions, so that every output frame is a probability distribution as required by Equation (4.1). The prior is also added to the auxiliary logits during training.

The design of the decoder follows from three considerations. First, the encoder carries most of the representational capacity, so the decoder can be small; its 7 million parameters are about 2 % of the ViT-L encoder. Second, the (2+1)D factorisation adds temporal mixing at every scale at a lower cost than full three-dimensional kernels [34]. Third, predicting maps for many frames in one forward pass amortises the cost of the encoder: a single pass over a 16-frame clip yields usable maps for 16 original frames (Section 4.7). The measured cost is reported in Section 5.5.4.

4.5. Components Evaluated but Not Retained

Four further components were implemented, tested under the rules of Table 4.1, and not retained. They are described here because their negative results are part of the contribution of the thesis.

4.5.1. Predictive Surprise from the V-JEPA 2 Predictor

The original proposal was built around the hypothesis that the prediction error of a self-supervised video model indicates where something unexpected happens, and therefore where people look, in the spirit of the Bayesian surprise account of attention [38]. V-JEPA 2 provides the means to test this, because its released checkpoint includes the predictor used during pre-training.

For a window of 16 sampled frames, the first eight frames (four tubelets) are the context and the last eight the target. The context is encoded on its own, so that no information from the target frames leaks into it, and the predictor maps the context tokens to predicted target tokens $\hat{z}_p$. The target tokens z_p are taken from an encoding of the full window. With LN a parameter-free layer normalisation over the channel dimension, the *raw surprise* of target token p is

$$e(p) = \frac{1}{D} \sum_{c=1}^{D} \left| \mathrm{LN}(\hat{z}_p)_c - \mathrm{LN}(z_p)_c \right|. \tag{4.8}$$

A *contextual* variant subtracts the error obtained when the context is replaced by the context of a different video, $e(p) - e_{\mathrm{wrong}}(p)$, so that only the part of the error that depends on the true past remains. Two reference maps tested whether surprise adds anything beyond simpler cues: *token atypicality*, the mean absolute deviation of each target token from the mean token of its frame, and *frame difference*, the absolute grey-level difference between consecutive frames pooled to the 16×16 token grid. All maps were upsampled to 640×360, blurred with a Gaussian of $\sigma = 16$ pixels and scored on the first frame of each target tubelet of the 60 holdout videos, at about 4 and 15 frames per second. A decoder module that injects the surprise map through zero-initialised feature-wise modulation was implemented but, following the outcome of the gate, never trained.

4.5.2. Low-Rank Adaptation of the Encoder

Low-rank adaptation (LoRA) [39] fine-tunes a frozen linear layer $W \in \mathbb{R}^{d_{\mathrm{out}} \times d_{\mathrm{in}}}$ by adding a trainable low-rank update,

$$W'x = Wx + \frac{\alpha}{r} BAx, \qquad A \in \mathbb{R}^{r \times d_{\mathrm{in}}},\ B \in \mathbb{R}^{d_{\mathrm{out}} \times r}, \tag{4.9}$$

with B initialised to zero so that training starts from the frozen model. LoRA with $r = \alpha = 16$ was applied to the query, key, value and output projections of the last twelve blocks of ViT-L

(48 layers, 1.57 million parameters). Each LoRA model was initialised from the frozen model of the same seed and trained for three epochs with learning rates of 10^{-4} for the LoRA weights and 10^{-5} for the decoder.

4.5.3. Audio Fusion

To test whether the soundtrack helps on the audio-visual datasets, an audio fusion module was added to three pyramid levels (P_8, P_{16a}, P_{16b}). For each tubelet, the audio vector a (the mean of the features of its two frames) is embedded as $e = \mathrm{GELU}(W_p\,\mathrm{LN}(a)) \in \mathbb{R}^{256}$. A pyramid level F is then modulated as

$$\tilde{F} = F \odot (1 + \gamma) + \beta + W_r\big(F \odot s\big), \qquad s(x) = \mathrm{sigmoid}\left(\frac{\langle W_q F(x),\, W_k e\rangle}{\sqrt{256}}\right), \tag{4.10}$$

where $(\gamma, \beta) = W_f e$ is a feature-wise linear modulation (FiLM) [40], $s(x)$ is an audio-visual correspondence map at each position x, and W_r is a $1 \times 1 \times 1$ convolution. Because W_f and W_r are initialised to zero, the module is exactly the visual model at the start of training. As a control, the trained model was also scored with the audio of each test video replaced by that of another test video ("rotated audio"); a model that uses sound should do better with the matched soundtrack.

4.5.4. Static-Image Co-Training

Training on static images helped early video saliency models [3]. To test whether it still helps on top of a video foundation encoder, the 10,000 SALICON training images [21] were added to each training epoch, each image repeated over the 16 input frames, with the same loss.

4.6. Training Procedure

Loss. The loss combines three standard saliency objectives: the Kullback–Leibler divergence of the prediction from the ground-truth density, the linear correlation coefficient, and the normalised scanpath saliency [2]. For a predicted distribution S, a target density Y and fixation counts F on the 64×64 grid,

$$\mathrm{KL}(Y \,\|\, S) = \sum_p Y(p)\big[\log(Y(p) + \epsilon) - \log S(p)\big], \tag{4.11}$$

$$\mathrm{CC}(S, Y) = \frac{\sum_p (S(p) - \bar{S})(Y(p) - \bar{Y})}{\big\|S - \bar{S}\big\|_2 \big\|Y - \bar{Y}\big\|_2 + \epsilon}, \tag{4.12}$$

$$\mathrm{NSS}(S, F) = \frac{1}{\sum_p F(p)} \sum_p F(p)\, \frac{S(p) - \bar{S}}{\mathrm{std}(S)}, \tag{4.13}$$

with $\epsilon = 10^{-8}$. The training loss over the central frames O of a clip is

$$\begin{aligned}\mathcal{L} = \frac{1}{|O|}\sum_{t\in O}\Big[\mathrm{KL}(Y_t\|S_t) + \big(1 - \mathrm{CC}(S_t, Y_t)\big) - 0.1\,\mathrm{NSS}(S_t, F_t)\Big] \\ + 0.25\sum_{a\in\{8,16\}}\frac{1}{|O|}\sum_{t\in O}\Big[\mathrm{KL}(Y_t\|S_t^a) + \big(1 - \mathrm{CC}(S_t^a, Y_t)\big)\Big],\end{aligned} \tag{4.14}$$

where S_t^a are the predictions of the auxiliary heads and the NSS term is averaged over frames with at least one fixation. The KL and CC terms follow the loss of SalFoM [7]; the small NSS term additionally rewards high predicted values at the fixated locations themselves, which the two distribution-based terms do not address directly. The loss is averaged over the clips of a batch.

Optimisation. The decoder was trained with AdamW [41] at a learning rate of 3×10^{-4} and weight decay 0.01; the prior used a learning rate of 10^{-3} without weight decay. The learning rate rose linearly during the first quarter epoch and then followed a cosine decay to zero [42]. Gradients were clipped to a norm of 1.0, and the forward pass ran in bfloat16 mixed precision. Each epoch drew eight random clips from every training video, and the batch size was eight. Clip start positions were drawn uniformly, allowing up to a quarter of the clip to extend beyond the start or end of the video, where frames are repeated. The only data augmentation was a horizontal flip of frames and targets with probability 0.5; the colour jitter mentioned in the original proposal was not implemented.

Schedule and model selection. The proposal foresaw 20 epochs. In the backbone comparison (Section 5.3), however, the holdout CC of every backbone peaked after the second or third epoch and then fell while the training CC kept rising. The schedule for all later runs was therefore fixed, before any of them started, at three epochs, with the checkpoint chosen by holdout CC after each epoch (computed on every fifth frame).

Fine-tuning on other datasets. For UCF Sports, the six audio-visual datasets and Hollywood-2 (Vig gaze), each model was initialised from the DHF1K model and trained for three further epochs with a learning rate of 10^{-4} (3×10^{-4} for the prior). The number of clips per training video was set so that an epoch contained about 1,500 clips (16 clips per video for UCF Sports). Each fold of the audio-visual datasets was trained and selected on its own holdout and its test videos were scored once.

Seeds and ensembles. The main models were trained with three seeds (0, 1 and 2). A three-seed ensemble averages the members' probability maps before post-processing. On the fine-tuning datasets, ensemble member k is fine-tuned from the DHF1K model of seed k. Table 4.3 summarises the settings.

Table 4.3: Training and inference settings of SURGE-D

Setting	Value
Input	16 frames, stride 2, 256×256, ImageNet normalisation
Encoder	V-JEPA 2 ViT-L/16 or ViT-g/16, frozen
Decoder channels	256 (head: 128)
Output	64×64 per frame, central 8 of 16 sampled frames
Loss weights	KL 1, CC 1, NSS 0.1, auxiliary 0.25
Optimiser	AdamW, weight decay 0.01 (prior: 0)
Learning rate (DHF1K)	3×10^{-4} (prior 10^{-3}), warm-up 0.25 epoch, cosine
Learning rate (fine-tuning)	10^{-4} (prior 3×10^{-4})
Epochs, batch size	3, 8
Clips per video and epoch	8 (DHF1K); about 1,500 clips per epoch elsewhere
Augmentation	horizontal flip ($p = 0.5$)
Precision	bfloat16 autocast, gradient clipping at 1.0
Selection	best epoch by holdout CC
Output blur σ	tuned on the holdout over $\{0, 4, 8, 12, 16, 24\}$ pixels
Seeds	0, 1, 2; ensemble = mean of probability maps

4.7. Inference and Post-Processing

A whole video is processed with overlapping windows (Algorithm 4.1). Successive windows advance by $(T/2)\, s = 16$ original frames, so the central sampled positions of consecutive windows tile the video without gaps or overlaps. An original frame that lies between two sampled positions receives the linear interpolation of their maps. The 64×64 map is upsampled bilinearly to the native resolution, smoothed with a Gaussian of standard deviation σ (in native pixels) and normalised to sum to one. The value of σ was chosen on the holdout by CC for each model and each ensemble, and the same map is then used for all metrics, as recommended by Kümmerer et al. [6].

Algorithm 4.1 Whole-video inference with SURGE-D

Require: frames $I_0, \ldots, I_{N-1}$; model f; clip length $T = 16$; stride $s = 2$; blur σ
Ensure: saliency maps $\hat{S}_0, \ldots, \hat{S}_{N-1}$ at native resolution

$h \leftarrow (T/2)\, s$; $\quad c \leftarrow (T/4)\, s$; $\quad b \leftarrow -c$ ▷ hop, offset of first central frame, window start
while $b + c < N$ **do**
  $n_k \leftarrow \min(\max(b + ks,\, 0),\, N - 1)$ for $k = 0, \ldots, T - 1$ ▷ clamp at the video ends
  $(S_0, \ldots, S_{T-1}) \leftarrow f(I_{n_0}, \ldots, I_{n_{T-1}})$ ▷ one forward pass, 64×64 maps
  for $m \leftarrow b + c$ **to** $\min(b + c + h,\, N) - 1$ **do**
    $q \leftarrow (m - b)/s$; $\quad j \leftarrow \lfloor q \rfloor$; $\quad w \leftarrow q - j$
    $M \leftarrow (1 - w)\, S_j + w\, S_{j+1}$ ▷ linear interpolation in time
    $\hat{S}_m \leftarrow \mathrm{normalise}\big(\mathrm{Gauss}_\sigma(\mathrm{upsample}_{H \times W}(M))\big)$
  end for
  $b \leftarrow b + h$
end while

For an ensemble, the members' maps M are averaged before upsampling and blurring. Horizontal-flip test-time augmentation was implemented but not used in any reported result.

4.8. Evaluation Metrics and Statistical Analysis

4.8.1. Metrics

Six metrics were computed, covering the two families distinguished by Bylinskii et al. [2]: distribution-based metrics, which compare the predicted map P with the ground-truth density Q, and location-based metrics, which evaluate P at the fixated pixels of F. All are computed at the native resolution of the ground truth. Let $z(P)$ denote P standardised to zero mean and unit standard deviation over the frame, $\hat{P} = P/\sum P$ and $\hat{Q} = Q/\sum Q$.

- **Correlation coefficient (CC)**, the Pearson correlation between P and Q, $\mathrm{CC} = \frac{1}{|\Omega|}\sum_p z(P)(p)\, z(Q)(p)$, where Ω is the set of pixels.
- **Normalised scanpath saliency (NSS)** [43], the mean standardised saliency at fixated pixels, $\mathrm{NSS} = \frac{1}{\sum_p F(p)}\sum_p F(p)\, z(P)(p)$.
- **Similarity (SIM)**, the histogram intersection $\sum_p \min(\hat{P}(p), \hat{Q}(p))$.
- **Kullback–Leibler divergence (KLD)**, $\sum_p \hat{Q}(p) \log\left(\varepsilon + \hat{Q}(p)/(\hat{P}(p) + \varepsilon)\right)$, with ε the machine epsilon of double precision as in the MIT benchmark code [2]; lower is better.
- **AUC-Judd (AUC-J)** [44], the area under the ROC curve obtained by thresholding P at the saliency values of the fixated pixels, with fixated pixels as positives and all other pixels as negatives. As in the reference implementation, a tiny random jitter breaks ties between quantised values.
- **Shuffled AUC (sAUC)** [45], [46], which takes its negatives from the fixations of other videos instead of from all pixels and therefore does not reward a centre bias. For each video, the fixations of one random frame of each of ten other videos of the same test set are pooled; for each frame, 100 random draws of as many negatives as there are fixations are scored with thresholds in steps of 0.1 and the areas are averaged.

Each metric is averaged over the frames of a video and then over the videos of the test set; on the audio-visual datasets, the dataset score is the mean over its folds. The shuffled AUC depends on the sampling of negatives, so its values are compared only within this thesis.

Validation of the evaluator. The metrics were implemented independently of any existing saliency code. Before any model of this thesis was scored, the evaluator was checked against reference code by re-scoring a fixed set of saliency maps (Section 5.2).

4.8.2. Statistical Analysis

When two models are scored on the same test videos, their difference is assessed with a paired bootstrap over videos [47]. For a metric m and videos $v = 1, \ldots, V$, let $d_v = m_A(v) - m_B(v)$. The mean difference $\bar{d}$ is reported with a 95 % percentile interval obtained from 10,000

resamples of the V videos with replacement. A model is said to be better on a metric only if the interval excludes zero. The bootstrap treats videos as the independent units and therefore accounts for the strong correlation between frames of the same video. On the audio-visual datasets, the test videos of all folds are pooled for this purpose, which is possible because each video is tested exactly once.

Published results of other methods were computed with other evaluation code and, sometimes, other protocol details. They cannot enter a paired test. They are reported side by side with the results of this thesis, and margins below 0.01 CC are treated as ties (Table 4.1).

4.9. Ethical Considerations

The thesis uses only eye-tracking data that were collected and released by other research groups under their own ethical procedures; no human participants were recruited, and no personal data beyond the anonymised gaze recordings and the public video content of these datasets were processed. The licences of the data and models were respected: the Vig gaze data and the V-JEPA 2 checkpoints carry permissive licences, and the benchmark videos were used for research only, as their distributors require.

Saliency prediction has dual-use potential. Models of where people look can improve video compression, accessibility and the evaluation of user interfaces, but they can also be used to design content that captures attention against the viewer's interest. The models of this thesis predict average gaze over groups of observers and contain no information about individuals; the work does not, however, remove the broader risk, and the released code is intended for research use.

Finally, the experiments consumed electricity on a single workstation graphics card. The design choices that keep the encoder frozen and amortise its cost over many output frames reduce this footprint compared with fully fine-tuned alternatives, and every result was obtained with a fixed number of runs decided in advance rather than by open-ended search.

CHAPTER FIVE

EXPERIMENTS, RESULTS AND DISCUSSION

Guide: Describe the setup, present the results objectively, compare them with existing methods, and then interpret them. The discussion should say what the results mean for each research question.

This chapter reports the experiments in the order in which they were run, because later experiments were planned on the basis of earlier outcomes. After the setup (Section 5.1) and the validation of the evaluation code (Section 5.2), the two preliminary gates are reported: the choice of the video encoder (Section 5.3) and the test of predictive surprise as a fixation cue (Section 5.4). The main results on DHF1K, the transfer to eight further test sets, qualitative examples and the computational cost follow in Section 5.5, and the ablations in Section 5.6. The chapter ends with a discussion organised by research question (Section 5.7) and an assessment of the threats to validity (Section 5.8). Unless stated otherwise, every number in this chapter was produced by the code and evaluator described in Chapter 4, and every test set was scored once per reported model.

5.1. Experimental Setup

All experiments ran on a single workstation (Table 5.1), one job at a time. A monitoring process checked the GPU temperature, hardware events, disk space and memory every minute and could stop training cleanly; it was never triggered.

Table 5.1: Experimental environment

Item	Specification
Graphics Processing Unit (GPU)	NVIDIA RTX A5000, 24 GB
CPU / RAM	Intel Xeon w7-2475X (20 cores) / 256 GB
Operating system	Windows 11 Pro
Language	Python 3.12
Framework	PyTorch 2.14 with CUDA 13.0 [48]
Model loading	Transformers 5.22 [28]
Other libraries	NumPy [49], OpenCV, SciPy

5.1.1. Implementation Details

The settings of Table 4.3 were used throughout. Training one three-epoch DHF1K model took about 15 minutes of training time with the ViT-L encoder (4.7 minutes per epoch of 4,320 clips) and about 31 minutes with ViT-g (9.9 minutes per epoch), plus the holdout evaluation

after each epoch. ViT-g fitted in 10 GB of GPU memory at a batch size of eight. In total, the experiments of this chapter comprise 13 DHF1K training runs for the gates, the main program and its ablations, 7 further DHF1K runs for the ViT-g, 600-video and 32-frame experiments, and 125 fine-tuning runs on the other datasets (including the 16 runs with audio). Holdout lists, configuration files and the log of every decision are part of the released code.

5.2. Validation of the Evaluator

Because the metrics were implemented from scratch, the evaluator was validated before any SURGE-D model was scored (Table 4.1). The reference maps were the DHF1K validation predictions of SalFoM [7], produced on the same machine with the authors' released weights, for which per-frame scores computed with the evaluation code released by the TMFI-Net authors [50] were available.

The evaluator reproduced the reference scores: CC 0.5639 against 0.5642 ($\Delta = -0.0003$) and NSS 3.3264 against 3.3268 ($\Delta = -0.0004$), well within the tolerance of the rule. The SIM, KLD and AUC-J values of the reference code use other normalisations and could not be compared directly; against a second, independently written evaluator that follows the MIT benchmark conventions, CC, NSS, SIM and KLD agreed per video to within 5×10^{-5} and AUC-J agreed in the mean. Two deviations from the reference procedures were found and corrected during this check, before any model was scored: the first AUC-J implementation lacked the tie-breaking jitter of the reference code (which lowered AUC-J by 0.010 on quantised maps), and the first shuffled AUC pooled all negatives instead of drawing them per split. With the corrected evaluator, SalFoM's validation maps score CC 0.5639, NSS 3.326, SIM 0.4130, KLD 1.297, AUC-J 0.9243 and sAUC 0.7406. These are the SalFoM numbers used in all comparisons below. They are slightly lower than SalFoM's published scores because they are computed on the validation set rather than on the official test set, on which the authors report CC 0.569 and NSS 3.353 [7].

5.3. Selection of the Video Encoder (Gate G1)

The proposal had chosen V-JEPA 2.1 [29], which extends the V-JEPA 2 objective with a loss on the visible tokens and with self-supervision at intermediate layers, specifically to obtain denser features. Its ViT-L model is distilled from a two-billion-parameter teacher and was cooled down at 384×384 pixels [29][1]. Gate G1 compared three ViT-L encoders under identical conditions: V-JEPA 2, V-JEPA 2.1 and UMT-L [51], the encoder family used by SalFoM. Each was frozen and trained with the same decoder for ten epochs with seed 0, all V-JEPA models at 256×256 pixels. UMT-L received 224×224 input because its position table is fixed to that size and to 16 frames.

[1] Checkpoint `vjepa2_1_vitl_dist_vitG_384` from the official repository.

Table 5.2: Gate G1: frozen ViT-L encoders with the same decoder, internal DHF1K holdout (60 videos, every fifth frame, no blur), best of ten epochs, seed 0

Encoder	Best epoch	CC↑	NSS↑	SIM↑	KLD↓
V-JEPA 2 ViT-L (256^2)	3	**0.5782**	**3.314**	**0.4432**	**1.218**
V-JEPA 2.1 ViT-L (256^2)	2	0.5692	3.295	0.4377	1.245
UMT-L (224^2) [51]	2	0.5464	3.101	0.4157	1.323

Table 5.2 shows that V-JEPA 2 led on all four metrics. V-JEPA 2.1 trailed by 0.009 CC, outside the margin of 0.005 within which the proposal's choice would have been kept, so V-JEPA 2 was adopted for all later experiments. UMT-L trailed by 0.032 CC. With one seed per encoder, part of the gap between the two V-JEPA variants may be seed noise; the gap to UMT-L is several times larger than the seed-to-seed spread observed later (0.003 CC). The ranking of the two V-JEPA variants runs against the expectation that denser features help a dense task. Two explanations are compatible with the data and were not separated: the decoder reads four intermediate layers, which may already supply the local detail that the last layer of V-JEPA 2 lacks, and the V-JEPA 2.1 model was used at a lower resolution than the one at which it was cooled down.

The learning curves in Figure 5.1 were the more consequential outcome of this gate. For every encoder, holdout CC peaked in the second or third epoch and then declined steadily, while training CC kept rising. A frozen encoder with a 7-million-parameter decoder thus overfits the 540 training videos after a few passes. On the basis of these curves, and before any further run, the schedule of the main program was shortened to three epochs (Section 4.6).

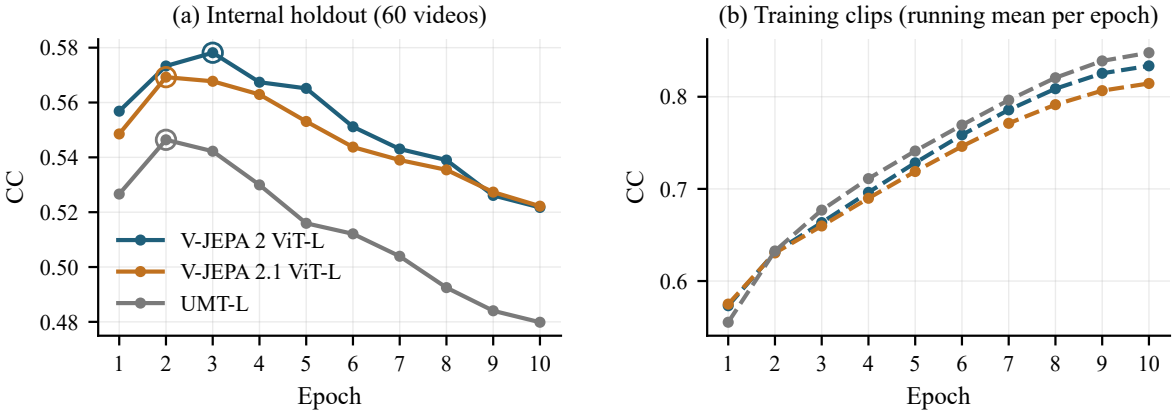


Figure 5.1: Learning curves of gate G1. (a) Holdout CC after each epoch; circles mark the best epoch. (b) Mean CC on the training clips of each epoch. Holdout accuracy peaks early and then declines while training accuracy keeps rising, for all three encoders

5.4. Predictive Surprise as a Fixation Cue (Gate G2)

Gate G2 tested the hypothesis that motivated the original proposal: that the error of V-JEPA 2's pretrained predictor indicates where people look (Section 4.5.1). No training is involved; the

surprise maps and the reference maps were scored directly against the fixations of the 60 holdout videos.

Table 5.3: Gate G2: zero-shot fixation prediction by surprise and reference maps on the DHF1K holdout (60 videos), at about 4 and 15 sampled frames per second

	NSS↑		CC↑	
Map	4 fps	15 fps	4 fps	15 fps
Raw surprise, Equation (4.8)	−0.190	−0.502	−0.043	−0.099
Contextual surprise (true − other past)	0.116	−0.366	0.018	−0.072
Token atypicality	0.454	0.650	0.081	0.123
Frame difference	0.810	0.803	0.149	0.148
Empirical centre prior	1.464	1.479	0.289	0.292

The outcome was unambiguous (Table 5.3). Neither surprise variant beat token atypicality or frame difference at either frame rate, and the raw surprise was *anti*-correlated with fixations: its NSS is negative, so observers looked on average at regions that the predictor found *easier* than average to predict. Information gain over the centre prior was negative for every map. The gate therefore failed, and the surprise module was never trained.

A preliminary diagnostic, run before the gate on eleven clips without gaze data (ten Kinetics validation clips [52] and one animated film trailer), helps to explain this result. When the true past of a clip was replaced by another clip's past, the mean prediction error rose in every clip (from 0.644 to 0.731 at about 4 fps), so the predictor does use the context. Yet the spatial pattern of the error correlated at $r \approx 0.60$ with token atypicality and at $r \approx 0.62$ with the error map computed from the *wrong* past, and it correlated negatively with a central Gaussian ($r \approx -0.11$). The error map therefore mainly reflects how unusual each target region is, largely independently of the past, and it is lower at the centre of the frame, where observers tend to look. The released training configuration of V-JEPA 2 offers a plausible reason: its masks span the whole clip in time[2], so the predictor was trained to fill in masked scene content rather than to extrapolate into the future. Its error is consequently not an expectation-violation signal in the sense of Bayesian surprise [38].

5.5. Results

5.5.1. Quantitative Results on DHF1K

The main program trained the frozen-encoder model (labelled B1 in the research log) with three seeds. The LoRA variant lost to it in all three seeds (Section 5.6), so the frozen model became the main model. After the ViT-g encoder passed its own gate, the same recipe was repeated with ViT-g. Table 5.4 reports all four configurations on the DHF1K validation set,

[2] `temporal_scale: [1.0, 1.0]` in the released pre-training configuration files of the official repository.

next to SalFoM scored by the same evaluator and to published validation results of other methods.

Table 5.4: Results on the DHF1K validation set (100 videos, 60,273 frames). The upper block was scored by the evaluator of this thesis; the lower block is copied from the literature and was computed with other evaluation code. Best value per column in bold

Model	CC↑	NSS↑	SIM↑	KLD↓	AUC-J↑	sAUC↑
SalFoM [7] (released weights)	0.5639	3.326	0.4130	1.297	0.9243	0.7406
SURGE-D ViT-L, single (σ = 12)	0.5836	3.349	0.4492	1.222	0.9283	0.7636
SURGE-D ViT-L, 3-seed ensemble (σ = 8)	0.5910	3.418	0.4486	1.187	0.9302	0.7614
SURGE-D ViT-g, single (σ = 8)	0.6003	3.457	0.4531	1.170	0.9308	**0.7709**
SURGE-D ViT-g, 3-seed ensemble (σ = 8)	**0.6029**	**3.486**	**0.4574**	**1.160**	**0.9314**	0.7680
Published, other evaluators (as reported in [53])						
TMFI-Net [50]	0.554	3.201	0.428	–	0.924	–
THTD-Net [54]	0.553	3.188	0.425	–	0.924	–
ViNet-E [53]	0.549	3.134	0.409	–	0.922	–
STSANet [55]	0.539	3.082	0.411	–	0.920	–
DiffSal [56]	0.533	3.066	0.405	–	0.918	–

Comparison with SalFoM. SalFoM is the closest prior work: it was the first video saliency model built on a video foundation encoder, it reported the best DHF1K test results among the methods it compared with [7], and its released weights allowed a comparison under an identical protocol. Table 5.5 gives the paired bootstrap intervals. The ViT-g model is better than SalFoM on all six metrics as a single model and as an ensemble, with every interval excluding zero; the CC gain of the single model is +0.036 [+0.026, +0.048]. The ViT-L ensemble is also better on all six metrics, and the single ViT-L model on five, its NSS difference not being significant. The per-video differences in Figure 5.2 show that the gain is broad rather than driven by a few videos: the ViT-g ensemble has the higher CC on 79 of the 100 videos.

Table 5.5: Paired bootstrap differences on DHF1K validation (100 videos, 10,000 resamples), with 95 % intervals; differences whose interval excludes zero are in bold. The intervals for all six metrics are given in Table A.1

Comparison	ΔCC [95 % CI]	ΔNSS [95 % CI]
ViT-L single − SalFoM	**+0.020** [+0.010, +0.031]	+0.023 [−0.050, +0.097]
ViT-L ensemble − SalFoM	**+0.027** [+0.017, +0.038]	**+0.091** [+0.022, +0.164]
ViT-g single − SalFoM	**+0.036** [+0.026, +0.048]	**+0.130** [+0.060, +0.205]
ViT-g ensemble − SalFoM	**+0.039** [+0.029, +0.050]	**+0.160** [+0.089, +0.233]
ViT-g single − ViT-L single	**+0.017** [+0.012, +0.022]	**+0.108** [+0.075, +0.142]
ViT-g ensemble − ViT-L ensemble	**+0.012** [+0.008, +0.016]	**+0.069** [+0.041, +0.097]

Two qualifications apply. First, SURGE-D uses frames on both sides of the predicted frame (Section 4.2), whereas SalFoM predicts the last frame of its input clip; part of the difference

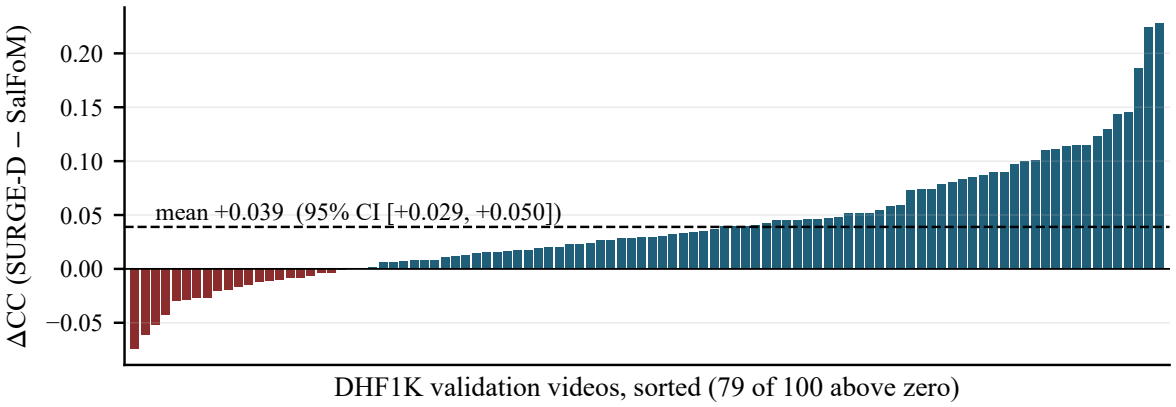


Figure 5.2: Per-video CC difference between the SURGE-D ViT-g ensemble and SalFoM on the 100 DHF1K validation videos, sorted. The dashed line is the mean difference; the interval is the paired bootstrap 95 % interval

may therefore come from access to future context, which SalFoM does not use. Second, the comparison involves one re-scored baseline. The published validation scores in the lower block of Table 5.4 were computed with other code, which agrees with the present evaluator to within 0.0003 CC on the reference maps (Section 5.2); with that caveat, SURGE-D's validation CC is 0.046–0.049 above the highest of them. Reported validation scores also differ between papers: UniST [57], for instance, lists TMFI-Net at 0.524 rather than 0.554 and reports 0.541 for itself. Several other peer-reviewed models published since 2024 [58], [59], [60] report DHF1K results whose tables could not be accessed or that refer to the test set; test-set scores cannot be compared with the validation scores of this thesis.

DHF1K test set. Maps for the 300 official test videos were produced with the ViT-g three-seed ensemble and checked for format by passing five validation videos through the same export, which reproduced their validation scores to within 10^{-4} CC. The scores on the test set are computed by the dataset authors on request and were not available when this thesis was written, so no test-set result is claimed.

5.5.2. Generalisation to Other Datasets

UCF Sports. UCF Sports differs from DHF1K in two ways: its gaze was recorded during an action-recognition task [10], and it is small (103 training videos). Without any UCF training, the DHF1K model reached a CC of 0.608 (Table 5.6), well below the fine-tuned models, which confirms that the task-driven gaze of this dataset differs from free viewing. After fine-tuning, the ViT-g ensemble reached CC 0.735 and the best NSS, SIM and AUC-J of the SURGE-D models.

On UCF Sports the encoder made a smaller difference than on DHF1K: the ViT-g ensemble was not significantly better than the ViT-L ensemble (ΔCC +0.0019 [−0.0071, +0.0125]), whereas the three-seed ViT-L ensemble was significantly better than its single member

Table 5.6: Results on the UCF Sports test set (47 videos, 2,929 frames). Published results were computed with other evaluation code

Model	CC↑	NSS↑	SIM↑	KLD↓	AUC-J↑	sAUC↑
SURGE-D ViT-L, zero-shot	0.6077	3.128	0.5033	1.113	0.9223	0.7453
SURGE-D ViT-L, fine-tuned	0.7252	3.916	0.5799	0.773	0.9406	0.8089
SURGE-D ViT-L, 3-seed ensemble	0.7331	4.028	0.5833	0.749	0.9430	0.8153
SURGE-D ViT-g, fine-tuned	0.7305	4.061	0.5831	0.747	0.9442	0.8145
SURGE-D ViT-g, 3-seed ensemble	0.7350	4.085	0.5857	**0.736**	**0.9449**	**0.8173**
Published ([53]; SalFoM from [7])						
ViNet-E [53]	**0.744**	**4.156**	**0.587**	–	0.941	–
ViNet-A [53]	0.734	4.108	0.586	–	0.940	–
THTD-Net [54]	0.711	3.840	0.565	–	0.933	–
TMFI-Net [50]	0.707	3.863	0.565	–	0.936	–
SalFoM [7]	0.631	3.543	0.516	–	0.928	–

(+0.0080 [+0.0039, +0.0123]). Against published results, SURGE-D clearly exceeds SalFoM (by about 0.10 CC) and TMFI-Net, ties ViNet-A (+0.001) and remains below ViNet-E (−0.009, a tie under the rule of Table 4.1, but below ViNet-E on CC, NSS and SIM, though above it on AUC-J). SURGE-D is therefore competitive on UCF Sports but not ahead of the best published model. Both ViNet variants use encoders pretrained for action recognition or action localisation [53], which may suit the task-driven gaze of this dataset.

Audio-visual datasets. Table 5.7 reports the six datasets of the STAViS benchmark [16], as fold means. The DHF1K model already transferred reasonably without any training (zero-shot row), and fine-tuning improved every dataset, most of all Coutrot2 (+0.17 CC), whose meetings differ most from DHF1K. With ViT-g, the ensemble was significantly better than the ViT-L ensemble on DIEM, Coutrot1, ETMD and AVAD (paired bootstrap over the pooled test videos), not significantly different on Coutrot2, and tied on SumMe.

Against the best peer-reviewed results that could be verified, the ViT-g ensemble has the higher CC on five of the six datasets. The margin is clear on DIEM, Coutrot1, ETMD and SumMe. On AVAD, the margin over the best visual-only model is at the threshold (+0.011), and the result ties the best audio-visual model. On Coutrot2 it is lower than both. The other metrics broadly agree with CC: the ensemble's NSS (DIEM 2.88, Coutrot1 3.30, ETMD 3.69, AVAD 4.31, SumMe 3.26) is above the published values of both reference models on these five datasets, while SIM is lower than DiffSal's on Coutrot1 and AVAD.

These comparisons have three limits. They cross evaluation code. They are restricted to peer-reviewed results, so results that had not appeared in peer-reviewed venues when this thesis was written are not included. And they are restricted to results whose published tables could be accessed: further recent audio-visual models, among them TM2SP [58], FOSal [62] and the model of Mao et al. [63], could not be checked and are therefore not represented. The table should consequently not be read as establishing a state of the art on these datasets.

Table 5.7: CC on the six audio-visual datasets (fold means; DIEM has one split). The lower block lists the best peer-reviewed results whose tables could be verified, computed with other evaluation code. Visual-only: ViNet-E, or ViNet-A on Coutrot2 [53]; on SumMe, which those models do not report, the visual-only variant of the relevance-guided fusion model AVRSP of Yu et al. [61]. Audio-visual: AVRSP [61] on DIEM and Coutrot2, DiffSal [56] on the other four

Model	DIEM	Coutrot1	Coutrot2	ETMD	AVAD	SumMe
SURGE-D ViT-L, zero-shot	0.675	0.600	0.645	0.588	0.692	0.551
SURGE-D ViT-L, fine-tuned	0.706	0.632	0.816	0.662	0.718	0.581
SURGE-D ViT-L, 3-seed ensemble	0.714	0.641	0.807	0.670	0.729	**0.593**
SURGE-D ViT-g, fine-tuned	0.714	0.648	0.820	0.672	0.731	0.586
SURGE-D ViT-g, 3-seed ensemble	**0.729**	**0.657**	0.819	**0.676**	**0.740**	0.590
Best visual-only (published)	0.701	0.614	**0.862**	0.632	0.729	0.494
Best audio-visual (published)	0.685	0.638	0.837	0.652	0.738	0.572
ViT-g ens. − best visual-only	+0.028	+0.043	−0.043	+0.044	+0.011	+0.096
ViT-g ens. − best audio-visual	+0.044	+0.019	−0.018	+0.024	+0.002	+0.018

Coutrot2 is discussed in Section 5.7.

Hollywood-2 with the Vig gaze data. Table 5.8 reports the 884 test clips of Hollywood-2 with the gaze of Vig et al. [12] (257,733 scored frames). These numbers are not comparable with published Hollywood-2 results (Section 4.3). With this many test clips, small differences are resolved precisely: ViT-g was better than ViT-L on all six metrics as a single model (ΔCC +0.0046 [+0.0032, +0.0060]) and on five as an ensemble (+0.0041 [+0.0030, +0.0052]). Ensembling raised CC, KLD, AUC-J and sAUC but significantly *lowered* NSS and SIM, for both encoders. With at most five observers per frame, the ground truth is sparse, and the averaging of three members smooths the peaks that NSS and SIM reward.

Table 5.8: Results on Hollywood-2 with the gaze of Vig et al. (884 test clips). Not comparable with published Hollywood-2 results, which use other ground truth

Model	CC↑	NSS↑	SIM↑	KLD↓	AUC-J↑	sAUC↑
SURGE-D ViT-L, zero-shot	0.632	3.595	0.503	1.061	0.951	0.780
SURGE-D ViT-L, fine-tuned	0.6798	3.898	0.5178	0.925	0.9564	0.8167
SURGE-D ViT-L, 3-seed ensemble	0.6841	3.875	0.5073	0.918	0.9574	0.8192
SURGE-D ViT-g, fine-tuned	0.6844	**3.933**	**0.5214**	0.917	0.9569	0.8180
SURGE-D ViT-g, 3-seed ensemble	**0.6882**	3.907	0.5104	**0.911**	**0.9577**	**0.8199**

5.5.3. Qualitative Results

Figure 5.3 shows frames from five DHF1K validation videos, chosen by a rule fixed before any map was inspected: the videos at the 10th, 50th and 90th percentile of the ensemble's per-video CC, and the videos with the largest gain and the largest loss relative to SalFoM; the frame shown is the middle frame of each video.



Figure 5.3: Qualitative examples on DHF1K validation. Rows: videos at the 10th, 50th and 90th percentile of SURGE-D's per-video CC, and the videos with the largest gain (0696) and loss (0663) relative to SalFoM; columns: middle frame with the observers' fixations, ground-truth density, SalFoM and the SURGE-D ViT-g ensemble, overlaid on the frame. Numbers give the CC of the shown frame. Video-level CC (SURGE-D / SalFoM): 0650 0.455/0.376, 0632 0.613/0.639, 0685 0.741/0.715, 0696 0.664/0.437, 0663 0.388/0.462

On these frames, the two models produce similar maps: both locate the horse rider, the surfer and the flying object that observers follow, and on the single frames shown SalFoM is in fact marginally closer to the ground truth in four of the five rows and equal in the fifth. Single frames are a poor guide to average differences that amount to a few hundredths of CC: for video 0696, for example, SURGE-D's CC over the whole video is 0.23 higher, yet the two maps at its middle frame are almost equal. The last row shows a characteristic failure. In a panel discussion, the observers' fixations concentrate on one person on the right, while SURGE-D spreads its prediction over several faces at the table. In conversation scenes, gaze is drawn to the faces of the people who are speaking [15], which a model that sees only the image cannot determine; the same limitation reappears on Coutrot2 below.

5.5.4. Computational Cost

Table 5.9 reports the cost of SURGE-D, measured with the PyTorch FLOP counter on one $16 \times 256 \times 256$ clip and timed on one DHF1K validation video with frames preloaded into memory (bfloat16, network forward pass only). Because one forward pass yields maps for 16 original frames, the cost per output frame is one sixteenth of the cost per clip.

Table 5.9: Computational cost of SURGE-D on an RTX A5000 (forward pass only). TFLOPs count multiply–add pairs as two operations

	ViT-L	**ViT-g**
Encoder parameters (frozen)	303.9 M	1,012.2 M
Decoder parameters (trained)	6.99 M	7.39 M
Encoder TFLOPs per clip	1.656	5.088
Decoder TFLOPs per clip	0.166	0.168
TFLOPs per output frame	0.114	0.328
Throughput, batch 4 (output frames/s)	301	129
Latency per clip, batch 1 (ms)	116	163
Peak GPU memory, batch 4 (GB)	2.8	7.1

Both models run faster than real time on this GPU. For comparison, the UMT-L encoder that SalFoM fine-tunes requires 1.90 TFLOPs for one $16 \times 224 \times 224$ clip (3,136 tokens; counted on a re-implementation loaded from the released checkpoint), and SalFoM runs it once per output frame in a sliding window [7]. Counting the encoder alone, SalFoM therefore needs about 17 times the computation per output frame of SURGE-D with ViT-L, and about 6 times that of SURGE-D with ViT-g. The multi-frame output thus allows an encoder of three times the size at a fraction of the cost of a per-frame design.

5.6. Ablation Study

Table 5.10 collects the ablations. Each was evaluated on the holdout under the rules of Table 4.1; for single-seed runs, the reference is the frozen ViT-L model of seed 0 trained with

the same three-epoch schedule.

Encoder adaptation. LoRA was expected to help, because adapting a pretrained encoder to the target task usually improves on frozen features, and LoRA does so while training only a small fraction of the parameters that full fine-tuning would change [39]. Here it did not. Each LoRA model started from the frozen model of the same seed and lost to it on the holdout in all three seeds (mean −0.0025 CC), and its best epoch was always the first. With only 540 training videos, adapting 1.6 million encoder parameters appears to add more variance than signal; the same early overfitting that limits the decoder (Figure 5.1) would act on the adapted encoder as well. Full fine-tuning, as done by SalFoM, was not tested.

Components. Removing the learned centre prior or the auxiliary heads, and adding 10,000 static SALICON images per epoch, changed holdout CC by less than the threshold of 0.003. The centre prior's lack of effect suggests that the encoder features and the convolutional decoder can already represent the centre bias, for instance through the image borders, which the squashed input keeps at fixed positions; the explicit prior then adds nothing that can be measured. The static images, which helped CNN-era models [3], appear redundant once the encoder has been pretrained on large amounts of video. A stride of four, which halves the cost per output frame, reached +0.0031, exactly at the threshold; with one seed, this is a candidate for future work rather than a finding.

Encoder scale. Replacing ViT-L by ViT-g, with every other setting unchanged, raised holdout CC for all three seeds (0.5866, 0.5867 and 0.5829 against 0.5772, 0.5801 and 0.5776) and validation CC by 0.017 for single models (Table 5.5). Scale was the largest single improvement found after the choice of encoder family.

Temporal context and training data. Doubling the clip to 32 frames (about 2.1 s) reached a holdout CC of 0.5902, 0.0002 below its gate, so it was not pursued; unlike the 16-frame runs, it was still improving at its last epoch, so a longer schedule might change this conclusion, but that would be a new experiment. Training on all 600 DHF1K training videos instead of 540 gave validation scores within noise of the 540-video models. Neither more context nor 11 % more videos made a measurable difference.

Ensembles and output blur. Averaging three seeds raised validation CC by 0.007 for ViT-L and by 0.003 for ViT-g (Table 5.4); the stronger encoder leaves less variance for the ensemble to remove. Figure 5.4 shows that the optimal blur is broad: holdout CC varied by at most 0.003 between $\sigma = 0$ and $\sigma = 12$ for every configuration. The ViT-L ensemble preferred less blur than its single model (8 against 12 pixels), consistent with the smoothing that averaging already provides.

Table 5.10: Ablations on the DHF1K holdout (60 videos) unless marked "val". ΔCC is relative to the reference in each block

Variant	CC	ΔCC	Outcome under the rule
Components (ViT-L, seed 0; reference CC 0.5772, NSS 3.329)			
Without centre prior	0.5772	−0.0000	no difference
Without auxiliary heads	0.5757	−0.0015	no difference
With SALICON co-training	0.5801	+0.0028	no difference
Stride 4 (≈7.5 fps, 2 s window)	0.5803	+0.0031	difference (at the threshold)
Encoder adaptation (ViT-L, three seeds)			
LoRA, seed 0 / 1 / 2	0.5743 / 0.5791 / 0.5739	−0.0029 / −0.0010 / −0.0037	rejected (lost 3 of 3)
Encoder scale (seed 0; gate 0.5833)			
ViT-g instead of ViT-L	0.5866	+0.0083 vs ViT-L mean	adopted
Clip length (ViT-g, seed 0; gate 0.5904)			
32 frames instead of 16	0.5902	+0.0036 vs seed 0	not pursued (gate missed by 0.0002)
Training data (ViT-g, val, three-seed ensembles)			
600 instead of 540 train-	0.6044	+0.0015 [−0.0006, +0.0037]	not adopted

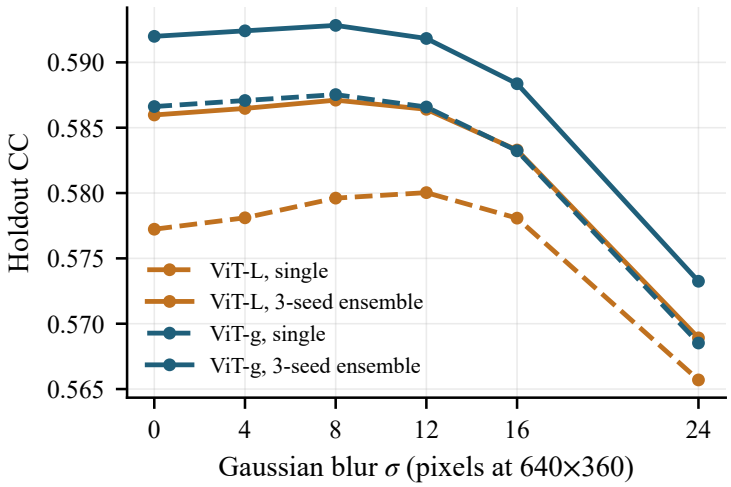


Figure 5.4: Holdout CC as a function of the Gaussian blur applied to the output, for single models and three-seed ensembles. The selected values are 12 (ViT-L single) and 8 (others)

Audio. The audio fusion module of Section 4.5.3 was trained on every fold of the six audio-visual datasets with the same recipe as the visual fine-tuning (ViT-L, seed 0). Table 5.11 shows that it did not improve the visual model on any dataset: the largest change was +0.001 CC, far below the threshold of +0.010. The control is informative nonetheless. On Coutrot2, scoring the trained model with the wrong soundtrack cost 0.018 CC, so the model had learned to use the audio there, but with the correct soundtrack it only matched the visual model, which already finds the faces in these meeting scenes from the image alone. With five test videos per fold, this observation is suggestive rather than conclusive.

Table 5.11: Audio fusion on the audio-visual datasets (ViT-L, fold-mean CC). The rule requires audio − visual ≥ +0.010 and matched − rotated ≥ +0.005

Dataset	Visual	Audio, matched	Audio, rotated	Audio − visual	Matched − rotated
DIEM	0.706	0.706	0.706	+0.0008	+0.0002
Coutrot1	0.632	0.633	0.630	+0.0010	+0.0031
Coutrot2	0.816	0.816	0.797	−0.0003	+0.0183
ETMD	0.662	0.661	0.661	−0.0014	−0.0001
AVAD	0.718	0.718	0.717	+0.0002	+0.0013
SumMe	0.581	0.579	0.580	−0.0021	−0.0011

5.7. Discussion

This section interprets the results in terms of the research questions of Section 1.4.

RQ1: Can a frozen self-supervised video encoder with a light decoder predict fixations better than the strongest foundation-model approach? On DHF1K, yes, within the limits of the available test data. With ViT-L, a frozen V-JEPA 2 encoder and a 7-million-parameter decoder already exceeded SalFoM, a model whose 300-million-parameter encoder is fully

fine-tuned, under an identical evaluation, on five of six metrics as a single model and on all six as an ensemble; with ViT-g, the margin of the single model grew to +0.036 CC and covered all six metrics. Two features of the result matter for its interpretation. First, the gain does not come from adapting the encoder: the frozen encoder outperformed its LoRA-adapted version, and the result therefore speaks to the quality of the pretrained V-JEPA representations rather than to a training technique. The gate G1 comparison, in which V-JEPA 2 outperformed UMT-L by 0.032 CC under the same frozen protocol, supports this reading, since UMT-L is the encoder family that SalFoM builds on. Second, the evidence is a validation-set comparison with one re-scored baseline and SURGE-D's access to future frames; the official test-set evaluation is still outstanding.

RQ2: Does predictive surprise carry information about where people look? Not in the form available from V-JEPA 2. Its predictor's error was anti-correlated with fixations at 15 fps and was beaten by both trivial reference cues. The diagnostics point to the reason: the error tracks the atypicality of each region rather than violated expectations, and the pre-training masks give the predictor no reason to forecast. This is a statement about one released predictor and one way of reading it out, not a refutation of surprise-based accounts of attention [38]; a predictor trained to extrapolate in time could behave differently.

RQ3: Which design factors matter when the encoder is frozen? The results separate the factors cleanly. The encoder dominates: its pre-training (V-JEPA 2 over V-JEPA 2.1 and UMT-L) and its scale (ViT-g over ViT-L) produced the only gains that were large relative to seed noise on DHF1K. Seed ensembles added a smaller but reliable gain. Nearly everything else, including LoRA, the learned centre prior, deep supervision, static-image co-training, 32-frame clips and 11 % more training videos, made no measurable difference. A consistent explanation is that, with a strong frozen encoder, the limiting factor is the small number of training videos: the decoder overfits within three epochs (Figure 5.1), and the additions that would increase its effective capacity do not help, whereas better features do.

RQ4: How well does the approach transfer, and does audio help? The model transferred well. Fine-tuned for three epochs, it matched or exceeded the best peer-reviewed visual-only and audio-visual results that could be verified on five of the six audio-visual datasets, and it was competitive on UCF Sports, where action-oriented encoders retain an edge. The one clear failure, Coutrot2, has a simple account: in recordings of conversations, observers look at the person who is speaking, which an image-only model cannot determine, as the last row of Figure 5.3 also illustrates. The audio experiment shows that adding the soundtrack in a simple way does not resolve this. The fusion module learned to depend on the audio on Coutrot2, but it did not improve on the visual model. This agrees with the observations of Jain et al. [64] and Agrawal et al. [65], who found that deep audio-visual saliency models tend to

become insensitive to their audio input and that none of a range of audio modules and fusion techniques improved on visual-only models across ten datasets. Exploiting sound in this setting probably requires explicit speaker localisation rather than global feature modulation.

Relation to the original proposal. The final model differs from the proposed one in three respects, each decided by a pre-registered test: the encoder is V-JEPA 2 rather than V-JEPA 2.1, it is frozen rather than adapted with LoRA, and the surprise branch that gave the project its name is absent. The proposal's ranking of what drives performance, which placed the encoder first, was confirmed. Its expectation that LoRA would help, based on results with much larger training sets, was not.

5.8. Threats to Validity

Internal validity. All selection used holdout data, every test set was scored once, and every rule was written in advance, which protects against selection bias. Two residual weaknesses remain. The component ablations used one seed, and they were judged against a fixed threshold rather than a statistical test, so effects of a few thousandths of CC cannot be excluded. In addition, several thresholds (for example the margin of 0.005 in the gates) were chosen by judgement from the observed seed spread rather than derived formally.

Construct validity. The six metrics capture different aspects of a saliency map [2] and did not always agree; ensembles, for instance, improved CC but lowered NSS and SIM on the sparse Hollywood-2 (Vig gaze) data. The use of one map for all metrics, with the blur chosen by CC, favours CC slightly over the other metrics.

External validity and comparability. Comparisons with published numbers cross evaluation code and, in places, protocol details, such as the handling of frames without fixations or the exact fold composition. Only one baseline, SalFoM, was re-scored under the protocol of this thesis. SURGE-D's predictions use about 0.27 s or more of future context, which causal models do not have. The DHF1K test-set result and the standard Hollywood-2 ground truth were not available. Finally, the pre-training data of V-JEPA 2 include large collections of internet video, and DHF1K is drawn from YouTube; an overlap between pre-training and test videos cannot be ruled out. The same is true for every method that relies on a large pretrained video encoder, including SalFoM.

CHAPTER SIX

CONCLUSION AND FUTURE WORK

6.1. Conclusion

> **Guide:** Summarise the problem, what was done and the main findings, and answer each research question briefly. Do not introduce new results here.

Write the conclusion here.

6.2. Contributions

Restate the main contributions of the thesis.

6.3. Future Work

> **Guide:** Describe the limitations of the work and the directions that follow from them.

Describe future work here.

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APPENDIX A

ADDITIONAL RESULTS

This appendix gives the complete statistics behind the summaries of Chapter 5. All values were produced by the evaluator of Section 4.8.

A.1. Paired Bootstrap on the DHF1K Validation Set

Tables A.1 and A.2 extend Table 5.5 to all six metrics. Each cell gives the mean per-video difference and, below it, its 95 % percentile interval over 10,000 bootstrap resamples of the 100 validation videos. For KLD, lower is better, so negative differences favour the first model.

Table A.1: Paired bootstrap differences between SURGE-D and SalFoM on DHF1K validation, all six metrics

Metric	ViT-L single	ViT-L ensemble	ViT-g single	ViT-g ensemble
CC	+0.0197 [+0.0095, +0.0305]	+0.0271 [+0.0171, +0.0376]	+0.0364 [+0.0259, +0.0475]	+0.0390 [+0.0286, +0.0498]
NSS	+0.0225 [−0.0499, +0.0967]	+0.0910 [+0.0218, +0.1637]	+0.1304 [+0.0603, +0.2047]	+0.1597 [+0.0891, +0.2333]
SIM	+0.0362 [+0.0291, +0.0436]	+0.0356 [+0.0285, +0.0430]	+0.0401 [+0.0328, +0.0476]	+0.0444 [+0.0371, +0.0520]
KLD	−0.0747 [−0.1021, −0.0486]	−0.1101 [−0.1359, −0.0858]	−0.1270 [−0.1527, −0.1022]	−0.1369 [−0.1634, −0.1119]
AUC-J	+0.0040 [+0.0022, +0.0060]	+0.0060 [+0.0042, +0.0079]	+0.0065 [+0.0048, +0.0084]	+0.0071 [+0.0053, +0.0090]
sAUC	+0.0230 [+0.0166, +0.0298]	+0.0208 [+0.0143, +0.0277]	+0.0304 [+0.0238, +0.0373]	+0.0275 [+0.0209, +0.0345]

A.2. All Metrics on the Audio-Visual Datasets

Table A.3 reports all six metrics of the SURGE-D ViT-g three-seed ensemble on the six audio-visual datasets, as fold means. The output blur was tuned on each fold's holdout; the selected values lay between 8 and 16 pixels.

A.3. Encoder Comparison on the Audio-Visual Datasets

Table A.4 gives the paired bootstrap comparison of the ViT-g and ViT-L ensembles on the audio-visual datasets. Because every video of these datasets is tested in exactly one fold, the test videos of all folds were pooled for the bootstrap.

Table A.2: Paired bootstrap differences between the ViT-g and ViT-L encoders on DHF1K validation, all six metrics

Metric	Single models	Three-seed ensembles
CC	+0.0167 [+0.0116, +0.0222]	+0.0118 [+0.0077, +0.0163]
NSS	+0.1079 [+0.0746, +0.1423]	+0.0687 [+0.0407, +0.0974]
SIM	+0.0039 [+0.0000, +0.0077]	+0.0088 [+0.0058, +0.0119]
KLD	−0.0523 [−0.0669, −0.0386]	−0.0268 [−0.0381, −0.0158]
AUC-J	+0.0025 [+0.0016, +0.0035]	+0.0011 [+0.0004, +0.0020]
sAUC	+0.0074 [+0.0041, +0.0107]	+0.0067 [+0.0038, +0.0095]

Table A.3: SURGE-D ViT-g three-seed ensemble on the audio-visual datasets (fold means; DIEM has one split)

Dataset	CC↑	NSS↑	SIM↑	KLD↓	AUC-J↑	sAUC↑
DIEM	0.729	2.878	0.584	0.664	0.923	0.735
Coutrot1	0.657	3.302	0.504	0.947	0.911	0.774
Coutrot2	0.819	6.673	0.601	0.683	0.966	0.759
ETMD	0.676	3.687	0.510	0.930	0.950	0.765
AVAD	0.740	4.306	0.565	0.795	0.942	0.782
SumMe	0.590	3.261	0.448	1.148	0.927	0.763

Table A.4: CC difference between the ViT-g and ViT-L three-seed ensembles on the audio-visual datasets (pooled test videos, 10,000 resamples)

Dataset	Test videos	ΔCC [95 % CI]	Reading
DIEM	17	+0.014 [+0.003, +0.027]	ViT-g better
Coutrot1	60	+0.017 [+0.005, +0.031]	ViT-g better
Coutrot2	15	+0.012 [−0.007, +0.032]	no significant difference
ETMD	12	+0.006 [+0.002, +0.010]	ViT-g better
AVAD	45	+0.011 [+0.005, +0.019]	ViT-g better
SumMe	23	−0.003 [−0.010, +0.004]	no significant difference

APPENDIX B

REPRODUCIBILITY AND SAMPLE CODE

The implementation was written for this thesis and does not import code from other saliency projects. This appendix lists its main parts, the commands that produce the main results, and the forward pass of the decoder.

B.1. Organisation of the Code

Table B.1 lists the main files. The research log records every decision rule with its date, written before the experiment it governs, and every result with the folder it was read from.

Table B.1: Main files of the implementation

File	Content
`surge/backbones.py`	frozen encoders (V-JEPA 2 ViT-L and ViT-g, V-JEPA 2.1, UMT-L) and their taps; predictor error for gate G2
`surge/model.py`	feature pyramid, (2+1)D decoder, auxiliary heads, centre prior, surprise and audio modules
`surge/data.py`	dataset readers, clip sampling, targets, inference windows
`surge/losses.py`	training loss (Equation (4.14))
`surge/metrics.py`	CC, NSS, SIM, KLD, AUC-J and sAUC
`surge/infer.py`	whole-video inference (Algorithm 4.1) and scoring
`surge/lora.py`	low-rank adaptation (Equation (4.9))
`scripts/train.py`	training with holdout selection and clean stop/resume
`scripts/evaluate.py`	scoring of models, ensembles or saved maps
`scripts/tune_sigma.py`	selection of the output blur on the holdout
`scripts/compare.py`	paired bootstrap over videos
`scripts/g2_surprise.py`	gate G2
`scripts/prep_hollywood_gaze.py`	Hollywood-2 (Vig gaze) maps
`scripts/prep_stavis_audio.py`	audio features for the audio experiment
`scripts/thesis_qualitative.py`	qualitative examples and cost measurements
`configs/`	configurations and all split and holdout lists
`research/log.md`	dated decision rules and results

B.2. Commands for the Main Results

The main DHF1K model with the ViT-g encoder (seed 0) is trained, its blur is tuned on the holdout, and it is scored once on the validation set with:

```
python scripts/train.py --config configs/main_b1.yaml --set \
    run_dir=runs/vitg_s0 train.seed=0 train.epochs=3 \
```

```
    train.warmup_epochs=0.25 model.backbone=vjepa2_g "data.size=[256,256]"
python scripts/tune_sigma.py --run runs/vitg_s0 \
    --out runs/vitg/sigma_single.json
python scripts/evaluate.py --run runs/vitg_s0 --split val --metrics all \
    --sigma 8 --out runs/vitg/val_single_s0.json
```

Passing `--run` three times (seeds 0, 1 and 2) to the last two commands evaluates the three-seed ensemble. The paired bootstrap against another evaluation file is computed with `python scripts/compare.py A.json B.json`.

B.3. Forward Pass of the Decoder

The listing below is the forward method of the decoder in `surge/model.py`, which implements Equations (4.3), (4.5) and (4.7) (the softmax over pixels is applied in the loss and at inference). The two branches used only by the audio and surprise modules are abbreviated.

```
def forward(self, taps, surprise=None, T_out=None, audio=None):
    p4, p8, p16a, p16b = self.fpn(taps)            # SimpleFPN-3D
    if self.audio:                                  # audio experiment only
        ...
    if self.use_scm:                                # surprise module (not used)
        ...
    d16 = self.r16(self.fuse16(torch.cat([p16a, p16b], 1)))
    d8 = self.r8(p8 + F.interpolate(d16, size=p8.shape[2:], mode="nearest"))
    d4 = self.r4(p4 + F.interpolate(d8, size=p4.shape[2:], mode="nearest"))
    out = {"logits": self.head(d4)[:, 0]}           # B, T', 64, 64
    if self.aux and self.training:
        out["aux16"] = self.aux16(d16)[:, 0]
        out["aux8"] = self.aux8(d8)[:, 0]
    for k in list(out):
        x = out[k]
        if k != "logits":                           # aux maps to 64 x 64
            x = F.interpolate(x[:, None], size=out["logits"].shape[1:],
                              mode="trilinear", align_corners=False)[:, 0]
        if T_out is not None and x.shape[1] != T_out:   # tubelets -> frames
            x = F.interpolate(x[:, None], size=(T_out, *x.shape[2:]),
                              mode="trilinear", align_corners=False)[:, 0]
        if self.prior is not None:                  # + log G_theta
            x = x + self.prior(x.shape[-2], x.shape[-1], x.device)
        out[k] = x
    return out
```